CMPG 325 — Computer Networks

Milestone 1: Client Design Review

Project ID: CMPG325-2026-086 | Client ID: CLI-086 | Student: Morake, TM (39510484)

1. Client Requirements

Client: Mpho HR Consulting (Klerksdorp): Professional Services industry.

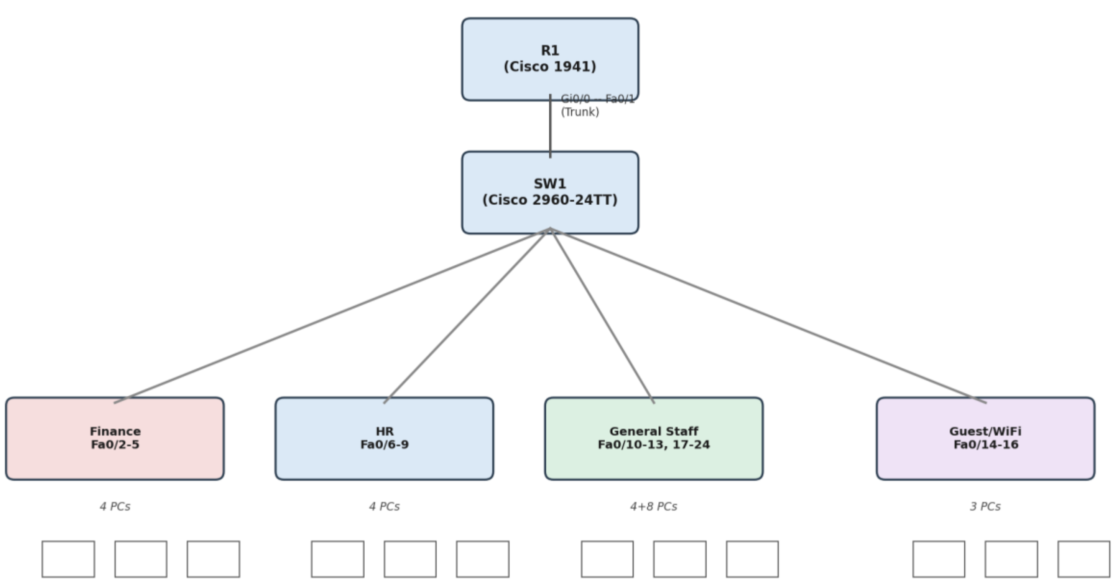
The client requires a segmented office network supporting four functional departments, with the following specific requirements drawn from the approved project brief:

- Assigned addressing block: 172.30.58.0/23, to be used as the basis for all department subnets.
- Provide appropriate connectivity and network services (DHCP, inter-VLAN routing) for all departments.
- Design constraint: The Finance department must be fully isolated from all general users no traffic may pass between Finance and any other department.
- Client Change Request (CR1): the client will sign 8 additional staff in one department. The network must accommodate this growth without any redesign (no new VLAN, no re-addressing, no topology change).
- Assigned technical challenge: VLANs (switch-based segmentation design), classified as Intermediate difficulty.
- Deliverables must be produced and demonstrated in Cisco Packet Tracer, with evidence maintained in an individual GitHub portfolio.

2. Physical Topology

The physical topology consists of a single Cisco 1941 router (R1) connected via a trunk link to a single Cisco 2960-24TT switch (SW1). All end-user devices connect to SW1 via individual access ports, grouped by department.

Mpho HR Consulting - Physical Topology

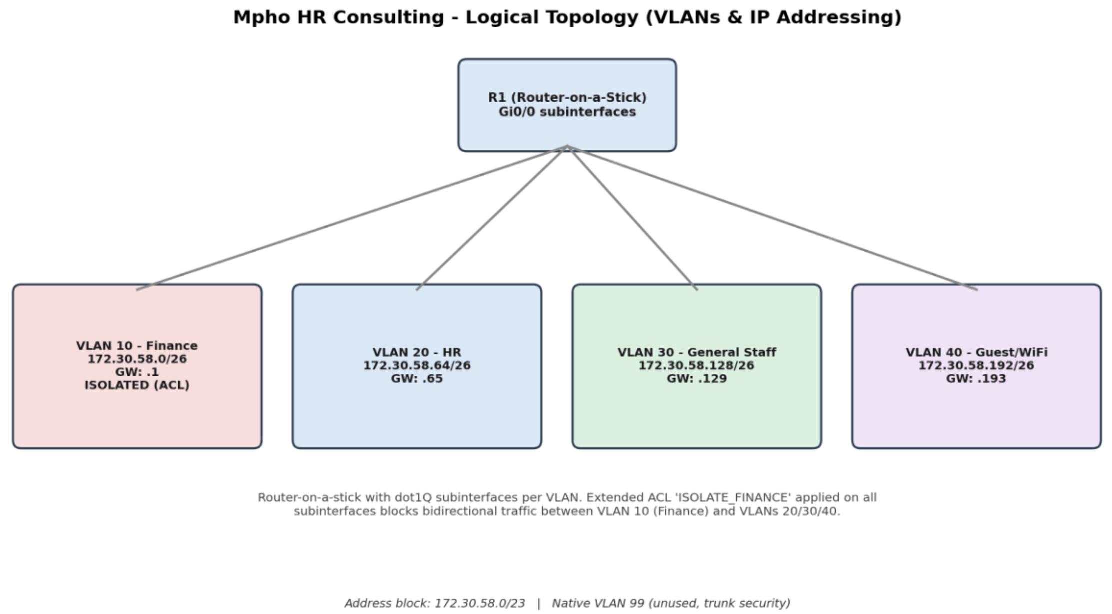


Single trunk link (router-on-a-stick) from R1 Gi0/0 to SW1 Fa0/1; all department PCs connect via access ports on SW1.

Department	Switch Ports	Device Count
Finance	Fa0/2 – Fa0/5	4 PCs
HR	Fa0/6 – Fa0/9	4 PCs
General Staff	Fa0/10 – Fa0/13 (+ Fa0/17–24 for CR1 growth)	4 PCs + 8 (CR1) = 12
Guest / WiFi	Fa0/14 – Fa0/16	3 PCs
Router uplink	Fa0/1 (trunk to R1 Gi0/0)	—

3. Logical Topology

Logically, the network uses a router-on-a-stick design: R1's single physical interface (Gi0/0) is divided into dot1Q subinterfaces, one per VLAN, each acting as that VLAN's default gateway. An extended access control list, ISOLATE_FINANCE, is applied inbound on every subinterface to enforce the Finance isolation constraint while leaving the other three departments free to communicate.



VLAN	Name	Subnet	Gateway
10	Finance	172.30.58.0/26	172.30.58.1
20	HR	172.30.58.64/26	172.30.58.65
30	General Staff	172.30.58.128/26	172.30.58.129
40	Guest / WiFi	172.30.58.192/26	172.30.58.193
99	Native (unused)	—	—

4. IP Addressing Plan

The assigned block 172.30.58.0/23 (172.30.58.0 – 172.30.59.255, 512 addresses) is subdivided into four /26 subnets (62 usable hosts each), one per VLAN. This sizing was chosen deliberately: each department currently uses roughly 4–15 hosts, leaving substantial headroom in every subnet. The second half of the block, 172.30.59.0/24, is left entirely unallocated for future departments or services.

VLAN / Dept.	Subnet	Usable Range	Broadcast	Gateway
10 Finance	172.30.58.0/26	.1 – .62	172.30.58.63	172.30.58.1
20 HR	172.30.58.64/26	.65 – .126	172.30.58.127	172.30.58.65
30 General Staff	172.30.58.128/26	.129 – .190	172.30.58.191	172.30.58.129
40 Guest/WiFi	172.30.58.192/26	.193 – .254	172.30.58.255	172.30.58.193

Why this plan satisfies the CR1 change request

General Staff (VLAN 30) was chosen as the department receiving 8 additional staff. Its /26 subnet provides 62 usable addresses against a starting population of 4 (demonstration) to a realistic 15–25 in practice adding 8 more hosts still leaves the subnet well under half full. Because addressing is handled by DHCP scoped to the existing subnet, new staff PCs are simply cabled to spare access ports already assigned to VLAN 30 and automatically receive an address from the same pool. No new VLAN, subnet, or router subinterface is required the change is absorbed entirely within the original design.

DHCP allocation

Each VLAN is served by its own DHCP pool on R1, scoped to that VLAN's subnet, with the first several addresses excluded (reserved for the gateway and any future static assignments).

Pool Name	Network	Default Router	Excluded Range
FINANCE_POOL	172.30.58.0/26	172.30.58.1	.1 – .10
HR_POOL	172.30.58.64/26	172.30.58.65	.65 – .70
GENERAL_STAFF_POOL	172.30.58.128/26	172.30.58.129	.129 – .135
GUEST_POOL	172.30.58.192/26	172.30.58.193	.193 – .198

5. Initial GitHub Repository

An individual GitHub repository has been created to host the project's portfolio of evidence, structured as follows (populated progressively as the project develops through Milestone 2 and final submission):

- README.md: project overview, client summary, and navigation to evidence
- /requirements: client requirements and this design review documentation
- /design: topology diagrams and IP addressing plan
- /packet-tracer: the .pkt file and configuration exports
- /testing: connectivity test screenshots and verification evidence
- /troubleshooting: notes on issues encountered and resolutions

https://github.com/Saint202/cmpg325-2026-086-mpho-hr-consulting